BOGA: A Brain-Inspired, Emotionally-Aware AI System for Context-Sensitive Interaction

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Abstract

Traditional chatbots and large language models (LLMs) are built to generate responses to every input, regardless of whether the input warrants one. While they may consider context or tone, they lack an internal mechanism for deciding when silence is more appropriate. In high-stakes environments such as courtrooms, such behavior can lead to distraction, inefficiency, and even procedural errors.

This paper introduces **BOGA**, a brain-inspired communication system that uses an emotional spike detection mechanism to determine when a language model should respond. Inspired by biological spiking neurons, BOGA remains silent until emotionally or contextually significant input is detected — mimicking the brain's ability to filter and time responses.

The architecture combines a **sentiment-based trigger** with a selectively activated LLM, aiming for more context-sensitive, efficient, and human-like interaction. A prototype implementation and example scenarios demonstrate the feasibility of this model for use in legal, therapeutic, and educational settings.

1. Introduction

In human conversation, we do not respond to every sentence we hear. We pause, we assess, and we speak when it matters. Most AI systems — particularly those powered by large language models — are trained to generate appropriate responses to a wide range of inputs. However, they **do not include a biologically inspired mechanism for deciding whether an input is emotionally or contextually *worth responding to at all.*** This missing layer of selectivity can result in excessive, unnecessary, or poorly timed interactions — especially in emotionally charged or high-stakes environments.

This paper proposes a novel approach to AI dialogue management inspired by the human brain. I introduce **BOGA**, an emotionally-aware AI system whose language model is only triggered by emotionally significant events, as detected by a specialized filtering mechanism. The core idea: instead of always responding, the AI will wait — and only speak when there is something worth saying.

2. Motivation

The inspiration for BOGA came from a deeply personal frustration with how we interact with AI today. Tools like ChatGPT are powerful, but they rely on *you* to remember to use them. You have to stop what you're doing, go to the tool, and then try to recall the exact details of what you wanted to ask. But often, by the time you're ready, the moment has passed — the idea is gone, or the emotion has faded.

This disconnect sparked a question:
"What if there was an AI that could stay with you — silently, intelligently — and capture those key moments as they happen?"

Rather than expecting humans to always initiate interaction, BOGA reverses the model. It **listens**, it **waits**, and it **responds only when necessary** — just like a human brain filtering important information from the noise of daily life.

The goal was to create an assistant that **acts like a second brain**: one that holds your passing thoughts, emotional moments, and context until you're ready to engage with it. BOGA isn't just about efficiency — it's about *preserving intelligence at the right moment*, when it matters most.

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3. Related Work

Recent years have seen growing attention to emotional AI, affective computing, and contextual filtering in dialogue systems. Sentiment analysis tools like VADER and TextBlob have been applied to classify emotions, and some chatbots integrate emotion into response generation. These systems are typically reactive, generating responses based on recognized sentiment — but they do not use biologically inspired timing or inhibition mechanisms to determine *whether* a response should be given at all.

Separately, **Spiking Neural Networks (SNNs)** have been studied as biologically realistic alternatives to traditional neural networks, particularly in neuroscience and robotics. Their event-driven, timing-sensitive nature makes them attractive for energy-efficient, brain-like computation. However, their integration into natural language systems remains largely unexplored. In particular, the use of SNNs to detect emotionally or contextually significant input and control the **timing of LLM activation** is not well-documented in existing literature.

To the best of my knowledge, no existing system combines:

- An **emotional spike detection mechanism**
- A **spiking neural network architecture**
- And a **gated LLM response flow** ...into a single AI assistant framework.

While related aspirations can be seen in global research trends — such as emotion-aware voice interfaces or energy-aware LLM gating — **BOGA's proposed integration appears novel**, particularly in how it mimics brain-like filtering and restraint. Additionally, no known equivalent system has emerged within Saudi Arabia's academic or technology ecosystem, making BOGA a potentially pioneering framework in the regional context.

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4. Proposed Method: The BOGA Architecture

BOGA is composed of two core components:

1. Emotional and Contextual Spike Detector:

To simulate the brain's ability to recognize emotionally or contextually significant moments, BOGA begins with a lightweight sentiment analysis component that assigns an intensity score to each line of input. If the score exceeds a certain threshold, it is treated as a "spike," signaling that the input may be meaningful enough to warrant a response.

While this initial approach allows for fast prototyping and testing of the system's gating behavior, it is limited in its depth and biological realism. In future iterations, this component will be replaced by a software-simulated spiking neural network (SNN) model, capable of more accurately capturing time-sensitive and event-driven neural behavior. This would enable BOGA to go beyond surface-level sentiment and incorporate other markers of significance — such as contradiction, urgency, or task-relevance — to reflect a more complete understanding of emotional and contextual salience.

2. Selective LLM Trigger:

The large language model (LLM), such as GPT or an open-source equivalent, remains inactive unless the spike detector identifies a significant emotional or contextual moment. Only then does the system generate a response. If no spike occurs, the model remains silent — a behavior inspired by the brain's natural inhibition and attention control mechanisms.

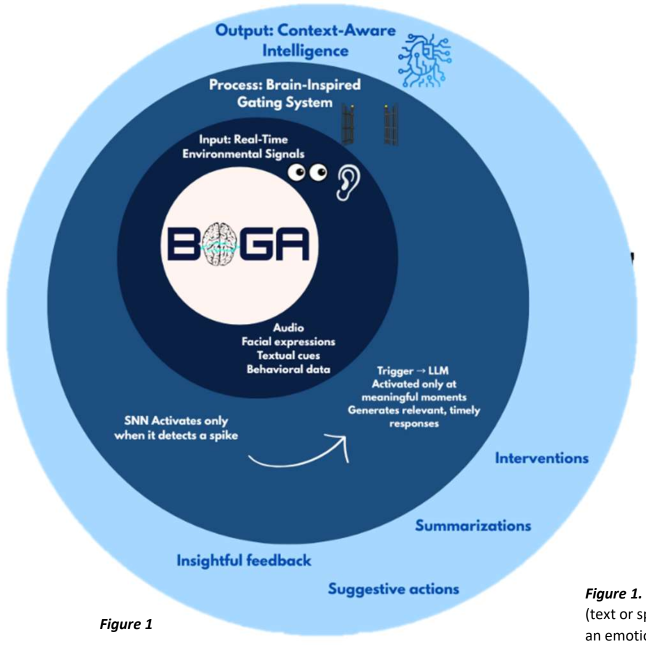

Figure 1

Figure 1. BOGA's architecture: Input (text or speech) is processed through an emotional or contextual spike detection mechanism. A response is generated only if the detected signal exceeds a defined threshold.

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5. Prototype Implementation

The current prototype of BOGA is implemented using a modular approach. Input is provided in the form of transcribed or typed text. An emotional spike detector processes this input by assigning intensity scores to each segment, which are compared against a defined threshold to determine whether to activate the language model. This approach allows for rapid testing of the system's core behavior: remaining silent during emotionally neutral input and speaking only when emotionally significant content is detected.

Although the architecture is designed to eventually handle real-time speech, this version focuses on text-based interactions to simplify early development. Future versions will incorporate automatic speech recognition (ASR) to support spoken dialogue, and will explore computational optimizations to enable efficient always-on performance.

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6. Potential Applications

BOGA can be applied to:

- **Legal environments**: real-time transcript filtering during trials
- **Mental health**: silent monitoring that only speaks during emotional spikes
- **Education**: more focused, less distracting AI in classroom settings
- **AI ethics research**: exploring the role of restraint and timing in AI communication

Its design emphasizes **efficiency**, **respect for silence**, and **emotional relevance** — unlike conventional chatbots.

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7. Future Work

Future versions of BOGA will:
• Replace basic sentiment thresholds with spiking neural network models for more biologically realistic emotional detection
• Expand into real-time audio input, not just text
• Test energy efficiency compared to conventional always-on LLM systems
• Explore the use of real-time brain signals (e.g., EEG or fNIRS) to trigger language model responses based on emotional or cognitive spikes. In this setup, **the brain signals do not inform the content of the response**, but rather serve as a timing mechanism — indicating that now may be a significant moment to initiate interaction.

This paper represents an early-stage proposal in a broader vision for brain-like, emotionally-aware communication systems.

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8. Author Note & Usage Disclaimer

This document presents an early-stage concept and prototype framework for **BOGA**, developed by **Escelle Mekhallalati** in April 2025.

The system is currently under development. Collaboration and academic feedback are welcome, but attribution is required for any referencing or derivative exploration.

9. References

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